

Unique completion, the lift, and the embeddability gap

Draft. Not a deposited preprint. Lean: lean/Full.lean

Abstract. A process function has one fixed point for every local intervention.

Checked: distinct outputs leave a party who disagrees in the output and agrees in w , and the source-and-sink extension is a bijection on bits.

Written: that witness kills the off-diagonal, so the induced map is unitary.

Not claimed: a metric, a wormhole Hamiltonian for Lugano, or the Leifer-Pusey no-go.

1. The demand

Faye, Stanford Encyclopedia, revised 28 October 2025, leaves the paradoxes open.

Dummett, 1954 and 1964, set the lexical objection aside. A stipulation is not a discovery.

Black, Analysis 1956, asked what happens to the intervention.

Lewis, 1976, split the modal and did not select the histories.

Horwich asked the theory to say what the intervention does.

Earman said a filter after the dynamics is not a dynamics.

No fixed point is the grandfather case. Two fixed points is the bootstrap.

The identity loop fails the clause. The swap fails it. A measure-zero rider cannot be failed.

2. Spacetime

Godel, 1949, has closed timelike curves. Morris, Thorne, Yurtsever, 1988, convert a wormhole.

Friedman et al., 1990, state the changed Cauchy problem.

Echeverria, Klinkhammer, Thorne, 1991, find one continuation, two, or none.

The billiard has the wrong multiplicity. Hawking's conjecture is a different question.

3. Quantum replies

Deutsch, 1991, asks for a fixed point of the partial trace. Uniqueness is not a theorem.

Aaronson and Watrous put the power at PSPACE. Bennett et al. call it an artefact.

Lloyd et al. replace the fixed point by post-selection. That is a simulation.

Oreshkov, Costa, Brukner drop global order. The quantum switch does not violate an inequality. Lugano does.

4. Equations

(1) $o = f(w(o))$

(2) $w(a,b,c) = (\text{not } b \text{ and } c, \text{ not } c \text{ and } a, \text{ not } a \text{ and } b)$

All 64 binary interventions have one solution. The image has four elements out of eight.

(3) U sends (o, e) to $(w(o)+e, o)$

(4) the inverse sends (i, s) to $(s, i+w(s))$

On bits the sum is exclusive or. Lean checked both compositions.

(5) if s differs from s -prime, some party disagrees in the output and agrees in w .

Otherwise one local function fixes both strings. Lean accepted this as `process_disagreement`.

(6) $\delta_k = w_k(s\text{-prime}) - w_k(s)$

(7) the gram entry is the product of row overlaps against that shift.

(8) $MM \text{ dagger} = I$. Written, not kernel-checked.

(9) a channel is $i_B = w(o_A)$, i_A independent of o_B .

(9) embeds as free propagation. (2) has no such wiring. The cyclic case stays open.

5. Lean

<https://github.com/drewarrowood/process-function-unitary-lift/blob/main/lean/Full.lean>

Checked there: the lemma and the bit bijection. Not checked: the gram identity.

6. References

Dummett, *Aristotelian Society Supp.* 28 (1954); *Philosophical Review* 73 (1964).

Black, *Analysis* 16 (1956). Lewis, *American Philosophical Quarterly* 13 (1976).

Horwich, *Asymmetries in Time* (1987). Earman, *Bangs, Crunches, Whimpers, and Shrieks* (1995).

Faye, *Backward Causation*, *Stanford Encyclopedia*, rev. 28 October 2025.

Godel, *Reviews of Modern Physics* 21 (1949).

Morris, Thorne, Yurtsever, *Physical Review Letters* 61 (1988).

Friedman et al., *Physical Review D* 42 (1990).

Echeverria, Klinkhammer, Thorne, *Physical Review D* 44 (1991).

Deutsch, *Physical Review D* 44 (1991). Aaronson and Watrous, *Proc. R. Soc. A* 465 (2009).

Bennett et al., *Physical Review Letters* 103 (2009). Lloyd et al., *PRL* 106 (2011).

Oreshkov, Costa, Brukner, *Nature Communications* 3 (2012).

Baumeler, Costa, Ralph, Wolf, Zych, *Classical and Quantum Gravity* 36 (2019).

Araujo et al., *Quantum* 1 (2017). Dourdent et al., arXiv:2609.39735.

Daher Ahmed and Kunjwal, arXiv:2610.00579. Leifer and Pusey, *Proc. R. Soc. A* 473 (2017).

Vilasini and Renner, arXiv:2203.11245. Stinespring, *Proc. AMS* 6 (1955).